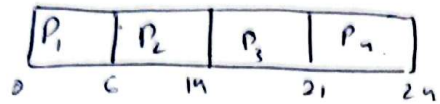


Lab 7.

Process	AT	BT	CT	TAT	WT
P_1	0	6	6	6	0
P_2	2	8	14	12	4
P_3	4	7	21	17	10
P_4	6	3	24	18	15

Gantt chart

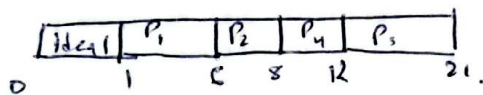


$$TAT = \frac{\sum T_i}{n} = \frac{6+12+17+18}{4} = 13.25 \text{ unit.}$$

$$\text{avg WT} = \frac{0+4+10+15}{4} = 7.25 \text{ unit.}$$

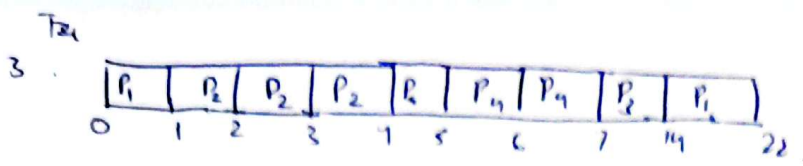
2 SJF.

Process	AT	BT	CT	TAT	WT
P_1	1	5	6	5	0
P_2	3	2	8	5	3
P_3	5	9	21	16	7
P_4	6	4	12	6	2



$$\text{avg tat} = \frac{5+5+16+6}{4} = 8 \text{ unit}$$

$$\text{avg WT} = \frac{0+3+7+2}{4} = 3 \text{ unit.}$$



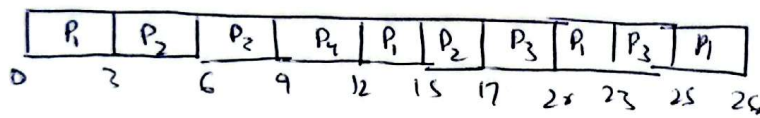
Process	AT	BT	CT	TAT	WT
P_1	0	9	22	22	13
P_2	1	4	5	4	0
P_3	2	7	14	12	5
P_4	3	2	7	4	2

avg tat = 10.5 unit

avg wt = 5 unit

4.

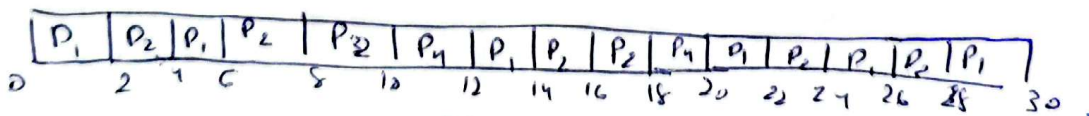
Process	AT	BT	CT	TAT	WT
P_1	0	10	26	26	16
P_2	1	5	17	16	11
P_3	2	8	25	23	15
P_4	3	3	12	7	6



avg tat = 18.5 unit

5.

Process	AT	BT	CT	TAT	WT
P_1	0	12	30	30	18
P_2	2	6	18	16	10
P_3	4	8	28	24	16
P_4	6	4	20	14	10



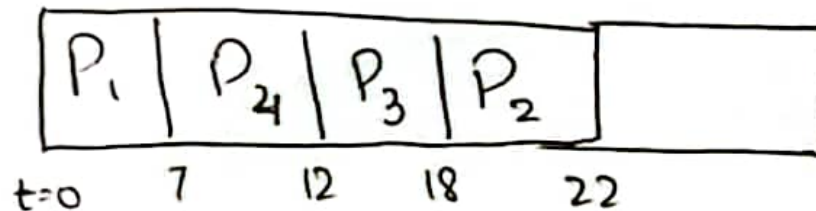
avg tat = 21 units

avg wt = 13.5 units

Task 6: Priority Scheduling

Process	AT	BT	Priority	CT	TAT	WT
P ₁	0	7	3	7	7	0
P ₂	1	4	1	20	21	17
P ₃	2	6	2	18	16	10
P ₄	3	5	4	12	9	5

Grant Chart.

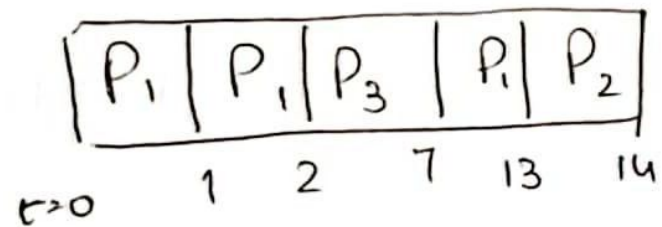


$$\text{avg TAT} = \frac{7+21+16+9}{4} = \frac{53}{4} = 13.25 \text{ unit} \quad \text{avg WT} = \frac{17+10+5+0}{4} = \frac{32}{4} = 8 \text{ unit}$$

Task 7:- Preemptive

Process	AT	BT	Priority	CT	TAT	WT
P_1	0	8	2	13	13	5
P_2	1	3	1	14	13	10
P_3	2	5	2	7	5	0

Gantt Chart:



$$\text{avg TAT} = \frac{13+13+5}{3} = \frac{31}{3} = 10.333 \text{ unit.}$$

$$\text{avg WT} = \frac{5+10+0}{3} = 5 \text{ unit.}$$